

Topic C8

The Periodic Table

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

Syllabus map

C8.1	Arrangement of elements
C8.2	Periodic trends
C8.3	Group I
C8.4	Group VII
C8.5	Transition elements
C8.6	Noble gases

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. Arrangement and electronic structure

Elements are arranged in increasing proton number. A period number gives the number of occupied electron shells; for main-group elements the group number relates to outer-shell electrons. Elements in the same group have similar chemical properties because they have the same number of valence electrons. Across a period, properties change from metallic to non-metallic.

Worked example

Electronic structure 2,8,7 identifies period 3, group VII (17), chlorine.

Examiner insight

Use proton number, not relative atomic mass, to explain modern ordering.

2. Group I: alkali metals

Lithium, sodium and potassium are soft, low-density metals forming +1 ions. They react with water to form metal hydroxide and hydrogen. Reactivity increases down the group because the outer electron is farther from the nucleus and more shielded, so attraction is weaker and electron loss is easier.



Examiner insight

Observations become more vigorous down the group; explanations must refer to electron loss.

3. Group VII: halogens

Chlorine, bromine and iodine are diatomic non-metals forming -1 ions. Colour darkens and volatility decreases down the group. Reactivity decreases because increased distance and shielding weaken attraction for an incoming electron. A more reactive halogen displaces a less reactive halide from solution.



Examiner insight

Displacement predictions compare the elemental halogen with the halide ion.

4. Transition elements

Transition elements are typical metals with high densities and melting points. They are less reactive than Group I, form coloured compounds, can have variable oxidation states and often act as catalysts. Iron uses +2 and +3 states; copper compounds are commonly blue or green.

Examiner insight

Give a named example when asked for a transition-element property, such as iron in the Haber process.

5. Noble gases

Group VIII/0 gases have full outer shells, are monatomic and chemically unreactive. Their boiling points increase down the group as atoms become larger and intermolecular attractions strengthen. Uses depend on inertness: argon in lamps or welding, helium in balloons.

Examiner insight

Noble gases are unreactive because of stable full outer electron shells, not because they contain no electrons.

Exam technique

High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

Practice Paper

Topic C8

he Periodic Table

Instructions

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Elements are arranged by increasing [1]
 - A nucleon number
 - B proton number
 - C density
 - D melting point

2. An element with electron arrangement 2,8,2 is in [1]
 - A period 2 group II
 - B period 3 group II
 - C period 2 group III
 - D period 3 group VIII

3. Group I metals form ions with charge [1]
 - A 1+
 - B 2+
 - C 1-
 - D 2-

4. Group I reactivity down the group [1]
 - A decreases
 - B increases
 - C is constant
 - D first rises then falls

5. Which halogen is most reactive? [1]
 - A iodine
 - B bromine
 - C chlorine
 - D astatine

6. Chlorine water added to potassium bromide forms [1]
 - A fluorine
 - B bromine
 - C iodine
 - D oxygen

7. Which is typical of transition elements? [1]
 - A only colourless compounds
 - B variable oxidation states
 - C very low density
 - D formation of -1 ions only

8. Which metal catalyses the Haber process? [1]
- A sodium
 - B iron
 - C calcium
 - D aluminium
9. Why are noble gases unreactive? [1]
- A They have no nucleus.
 - B Their outer shells are full.
 - C They are all liquids.
 - D They gain protons easily.
10. Boiling points of noble gases increase down the group because [1]
- A atoms become smaller
 - B interatomic forces weaken
 - C atoms become larger and attractions strengthen
 - D they become ionic

Section B · Structured questions

11 An element has proton number 16 and electron arrangement 2,8,6.

(a) State its period and group. [2]

(b) Predict the ion it forms. [2]

(c) Explain why it has similar chemistry to oxygen. [2]

(d) State whether it is metal or non-metal and justify. [2]

[Total: 8]

12 Sodium and potassium react with water.

(a) State two observations common to both. [2]

(b) Write the equation for sodium. [2]

(c) State which reacts more vigorously. [1]

(d) Explain the trend down Group I. [3]

[Total: 8]

13 Chlorine water is added separately to potassium bromide and potassium iodide.

(a) State the products in each tube. [2]

(b) Write the ionic equation with bromide. [2]

(c) Explain why chlorine reacts. [2]

(d) Predict whether bromine displaces chloride and explain. [2]

[Total: 8]

14 Compare iron with sodium.

(a) State two physical properties typical of transition metals. [2]

(b) State two chemical properties of transition elements. [2]

(c) Explain one catalytic use of iron. [2]

(d) Explain why sodium is stored under oil. [2]

[Total: 8]

15 Helium, neon and argon are noble gases.

(a) Explain why they are monatomic. [2]

(b) Explain their low reactivity. [2]

(c) Describe and explain the boiling-point trend. [3]

(d) State one use linked to inertness. [1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

Solutions · Multiple choice

- 1 B. Proton number defines the periodic sequence.
- 2 B. Three shells and two outer electrons.
- 3 A. They lose one valence electron.
- 4 B. The outer electron is lost more easily.
- 5 C. Reactivity decreases down Group VII.
- 6 B. Chlorine displaces bromide.
- 7 B. Transition metals commonly have variable oxidation states.
- 8 B. Finely divided iron is used.
- 9 B. A full valence shell is stable.
- 10 C. Larger electron clouds give stronger dispersion forces.

Solutions · Structured questions

11. 11(a) Period 3, Group VI/16 [2]. (b) S²⁻ [2]. (c) Both have six outer electrons [2]. (d) Non-metal; tends to gain/share electrons [2].

12. 12(a) Float/move, fizz, disappear, alkaline solution, heat [any 2]. (b) $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ [2]. (c) Potassium [1]. (d) More shells/shielding and distance weaken attraction, so electron lost more easily [3].

13. 13(a) Bromine and iodine respectively [2]. (b) $\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$ [2]. (c) Chlorine is more reactive / gains electrons more readily [2]. (d) No; bromine is less reactive than chlorine [2].

14. 14(a) High density, high melting point, strong/hard [any 2]. (b) Coloured compounds, variable oxidation states, catalysts [any 2]. (c) Iron catalyses Haber process by lowering activation energy [2]. (d) Prevents contact with oxygen/water because sodium reacts rapidly [2].

15. 15(a) Stable atoms do not need to bond [2]. (b) Full outer electron shell [2]. (c) Boiling point increases; atoms/electron clouds get larger and attractions strengthen [3]. (d) Argon in welding/lamp or another linked use [1].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.